

1. Administrative & Study Identification

Full Study Title: A Study Evaluating the Safety, Efficacy, and Pharmacokinetics of Mosunetuzumab and a Combined Regimen of Mosunetuzumab and Venetoclax in Participants With Relapsed or Refractory Chronic Lymphocytic Leukemia **Official Title:** A Phase IB Open-Label, Multicenter Study Evaluating the Safety, Efficacy, and Pharmacokinetics of Mosunetuzumab and a Combined Regimen of Mosunetuzumab and Venetoclax in Patients With Relapsed or Refractory Chronic Lymphocytic Leukemia **Protocol Number:** B043243 / 2022-501876-24-00 **NCT Number:** NCT05091424 **Trial Status:** Recruiting **Sponsor Name:** Roche (Hoffmann-La Roche) **Investigational Product:** Mosunetuzumab (Lunsumio), Venetoclax (Venclexta/Venclyxto), and Tocilizumab **Phase of Development:** Phase 1 (Phase 1b) **Medical Monitor:** Hoffmann-La Roche Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, pharmacokinetics, and preliminary efficacy of mosunetuzumab monotherapy and mosunetuzumab in combination with venetoclax in participants with relapsed or refractory (R/R) chronic lymphocytic leukemia (CLL). **Secondary Objectives:** To evaluate the objective response rate (ORR), the proportion of participants with undetectable minimal residual disease (MRD) in blood or bone marrow, and progression-free survival (PFS).

2.2 Background & Rationale Chronic lymphocytic leukemia (CLL) is a type of blood cancer where excess B-cells build up in the blood, bone marrow, and lymph nodes. Standard treatments include Bruton’s tyrosine kinase inhibitors (BTKi) and BCL2 inhibitors (like venetoclax). However, there is a significant unmet need for patients whose CLL has relapsed or become refractory to these standard therapies. Mosunetuzumab is an investigational CD20xCD3 bispecific T-cell engager that brings CLL cells and T cells together, directing the immune system to eliminate the cancer cells. This study evaluates mosunetuzumab alone and in combination with venetoclax to assess its safety and potential to provide deeper, more durable responses for this high-risk population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Multiple parallel cohorts, no placebo control)

Blinding: Open-label **Randomization:** Non-Randomized

3.2 Planned Enrollment & Duration Target N\$: 137 participants

Number of Sites: Multicenter (US, Europe, and Rest of World)

Estimated Study Start: Early 2022 **Estimated Primary Completion:** 31-May-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult men or women, ≥ 18 years of age. 2. Diagnosis of CLL requiring treatment according to the International Workshop on CLL (iwCLL) criteria. 3. Relapsed or refractory (R/R) CLL after failing at least two prior lines of therapy, with prior exposure to BTKi and/or venetoclax. 4. Eastern Cooperative Oncology Group (ECOG) performance score ≤ 2 . 5. Adequate bone marrow and liver function. 6. Life expectancy > 6 months.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding, or intending to become pregnant during the study or within 3 months after the final dose. 2. Prior treatment with mosunetuzumab or other CD20/CD3-directed bispecific antibodies; or prior allogeneic stem cell transplant. 3. Uncontrolled autoimmune hemolytic anemia or immune thrombocytopenia. 4. Known active infection with HIV, HTLV1, or SARS-CoV-2. (*Note: Arm C excludes patients who received venetoclax therapy within 12 months prior to first study treatment*).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm A:**

Mosunetuzumab subcutaneous (SC) given in increasing step-up doses weekly until the target dose is reached, then every 3 weeks for approximately 1 year. * **Arm B:** Mosunetuzumab SC (same schedule as Arm A) with continued BTKi overlap therapy for up to 6 weeks from the start of mosunetuzumab. * **Arm C:** Mosunetuzumab SC combined with daily oral venetoclax. * **Rescue:** Tocilizumab Intravenous (IV) as needed to manage cytokine release syndrome (CRS) events.

Route of Administration: Subcutaneous (mosunetuzumab), Oral (venetoclax, BTKi), Intravenous (tocilizumab). **Duration of Treatment:** Up to approximately 1 year (or until disease progression or unacceptable toxicity).

5.2 Concomitant & Prohibited Therapy Permitted: Tocilizumab for CRS management. Continuing BTKi therapy is permitted during the overlap phase for Arm B. **Prohibited:** Strong and moderate CYP3A inhibitors/inducers within 7 days prior to treatment; steroid

therapy for anti-neoplastic intent within 7 days. For participants receiving venetoclax, consumption of grapefruit, Seville oranges, or star fruit is prohibited.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Days -28 to 0	Informed Consent, Medical History, Confirmation of R/R status, Laboratory Tests
Baseline	Day 1	First Dose (Step-up dosing initiation)
Treatment Phase	Weekly (initial), then Q3W	Weekly visits during dose escalation to monitor for CRS; every 3 weeks for subsequent cycles. Efficacy and safety assessments
End of Study / Follow-up	Days & Long-term	End of treatment visit 1 month after the final dose; follow-up every 3 months for 2 years, then every 6 months to monitor for disease progression

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Rate of Dose-Limiting Toxicities (DLTs) from the first dose up to approximately 12 months (Arms A and B) or 24 months (Arm C). **Measurement Method:** Clinical observation, vital signs, and laboratory assessments evaluated according to standard toxicity criteria for dose escalation/safety confirmation.

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR) * Progression-Free Survival (PFS) * Minimal Residual Disease (MRD) Response Rate (Proportion of participants achieving MRD negativity in blood and bone marrow)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring during the step-up dosing phase (weekly hospital visits) for cytokine release syndrome (CRS) and neurologic toxicities. Standard AE collection occurs at every 3-week interval during maintenance. **Serious Adverse Events (SAEs):** Evaluated thoroughly for dose-limiting toxicity (DLT) determination, with provisions to administer IV tocilizumab immediately for CRS management. **Data Monitoring Committee (DMC):** Internal safety review committees oversee dose escalation rules and cohort progression.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The Safety Set will include all participants who receive at least one dose of study medication, used to evaluate DLTs and AEs. Efficacy analyses will be

performed on participants with measurable disease at baseline. **Sample Size Justification:** A target enrollment of 137 participants provides adequate statistical power to evaluate the safety profile (MTD/recommended dose) and preliminary efficacy signals in the R/R CLL cohorts. **Handling of Missing Data:** Standard Phase I missing data handling procedures; patients dropping out prior to DLT evaluation may be replaced.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of investigator sites by the sponsor to assure protocol adherence, safety reporting, and data integrity. **Audit Trail:** Validated Electronic Data Capture (EDC) systems will be used, alongside standard data clarification and source data verification workflows.

11. Study Identifiers & Governance

Primary ID: B043243 **Registry Number:** NCT05091424 **Sponsor/Lead Organization:** Roche **Study Title:** A Study Evaluating the Safety, Efficacy, and Pharmacokinetics of Mosunetuzumab and a Combined Regimen of Mosunetuzumab and Venetoclax in Participants With Relapsed or Refractory Chronic Lymphocytic Leukemia **Official Regulatory Title:** A Phase IB Open-Label, Multicenter Study Evaluating the Safety, Efficacy, and Pharmacokinetics of Mosunetuzumab and a Combined Regimen of Mosunetuzumab and Venetoclax in Patients With Relapsed or Refractory Chronic Lymphocytic Leukemia

12. Clinical Framework (CLL Specific)

Primary Condition: Relapsed or Refractory Chronic Lymphocytic Leukemia (R/R CLL) **Diagnosis Threshold:** Clinically confirmed CLL requiring treatment (iwCLL criteria) after failing ≥ 2 prior therapies **Medical Methodology:** Bispecific T-cell Engager (TCE) immunotherapy (alone or with BCL2 inhibition) **Optimization Target:** Eradication of leukemic clones / MRD negativity

13. Study Procedures & Methodology

Management Pathway: Step-up dosing schedule for mosunetuzumab to mitigate CRS risk **Education Protocol (HCP):** Site initiation and continuous training on CRS/neurotoxicity management and tocilizumab administration **Education Protocol (Patient):** Education regarding CRS symptoms, required hospital monitoring

schedules, and drug-drug/food interactions (e.g., avoiding grapefruit for venetoclax cohort) **Quality Audit Mechanism:** Real-time safety data review for DLTs and dose-escalation clearance

14. Observation & Follow-Up Schedule

Total Duration: Up to 24+ months (treatment and follow-up) **Onsite Visit Cadence:** Weekly during initial dose step-up, then every 3 weeks **Remote Monitoring Frequency:** Between cycles as needed per local site policies. **Checklist Review Interval:** Routine adverse event and concomitant medication checks at every cycle visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				